SSDU-style parameter tuning for noisy PCM

PnP-ADMM-DRUNet and GS-PnP-GS-DRUNet

Experiment report Version 3.0 8 November 2026

Parameter search: completed and frozen 450-image evaluation: complete

1 Executive summary

This experiment selects plug-and-play (PnP) reconstruction parameters for noisy photocurrent mapping (PCM) *without ground-truth images*. The previously reported "oracle-tuned" PnP baselines used ground-truth reconstruction scores on *validation* images, not individual test images. We replace that supervision with a measurement-only selection loss: each candidate reconstructs from part of the acquired Hadamard measurements and predicts the withheld part. This is inspired by self-supervised learning via data undersampling (SSDU) [1], but it tunes reconstruction hyperparameters rather than neural-network weights.

We retain four distinct selections: **PnP-ADMM-tuned-5**, **PnP-ADMM-tuned-50**, **GS-PnP-tuned-5** and **GS-PnP-tuned-50**. The 5-image subset contains one validation image from each of the five PCM categories; the 50-image subset contains ten per category. All 15 ADMM and all 18 GS-PnP configurations were evaluated on both subsets, and all selections were frozen. The runner reported **no candidate failures**. The 50-image parameter search completed on tuni on **8 November 2026 at 13:42:43 UK time**.

Method	Images	Candidate	Chosen parameters	SSDU loss
PnP-ADMM	5	`admm_009`	$\rho = 0.5$, $\sigma = 0.075$	0.00430443
PnP-ADMM	50	`admm_009`	$\rho = 0.5$, $\sigma = 0.075$	0.00451722
GS-PnP	5	`gs_018`	$\lambda = 0.1$, $\sigma = 0.075$, $\tau_0 = 33.3333$	0.00436543
GS-PnP	50	`gs_018`	$\lambda = 0.1$, $\sigma = 0.075$, $\tau_0 = 33.3333$	0.00443193

Table 1: Frozen results of the 5-image and 50-image tuning. SSDU loss is held-out measurement mean-squared error, *not* image MSE. The losses use different validation samples.

The two validation sizes selected **the same parameters** for each PnP algorithm, within the fixed search grids. This is evidence of selection stability, not by itself of high reconstruction quality. The **450-image test-set evaluation is complete**; Section 5 compares the frozen choices with their supervised-validation baselines using PSNR, SSIM, HaarPSI and MSE.

2 Tuning algorithm

Acquisition and four-fold splitting

For a noisy PCM acquisition, write

$$y = Ax + \varepsilon, \qquad n = 256^2, \qquad m = 13{,}107. \tag{1}$$

Here A is the *normalised* dyadic Hadamard sampling operator ($J = 8$) with 20% multilevel sampling and a fully included coarse block ($J_c = 6$). We use one archived NPL-default noisy acquisition per validation image; the original measurements are stored in float64 precision, while existing PnP solvers use float32 where necessary.

For each image i, keep all **4,096 coarse measurements** in the reconstruction set. Split only the other **9,011 acquired fine-scale measurements** into four deterministic, level-stratified held-out folds Λ_{ik} of approximately 2,253 measurements each (blocks of 128 row positions). Let Θ_{ik} contain all acquired measurements except the held-out fold. Thus $\Theta_{ik} \cap \Lambda_{ik} = \varnothing$. The held-out measurements are not passed to the solver.

Prediction loss and parameter selection

For candidate parameters θ, reconstruct using the fit measurements:

$$\widehat{x}_{\theta,ik} = \mathcal{R}_\theta(A_{\Theta_{ik}}, y_{\Theta_{ik}}). \tag{2}$$

Then calculate the loss *only on the withheld measurements*:

$$\ell_{ik}(\theta) = \frac{1}{|\Lambda_{ik}|} \left\| A_{\Lambda_{ik}} \widehat{x}_{\theta,ik} - y_{\Lambda_{ik}} \right\|_2^2 . \tag{3}$$

With S_c the images belonging to category c, the selection criterion is

$$\boxed{L_S(\theta) = \frac{1}{5} \sum_{c=1}^{5} \frac{1}{|S_c|} \sum_{i \in S_c} \frac{1}{4} \sum_{k=1}^{4} \ell_{ik}(\theta), \qquad \widehat{\theta}_S = \arg\min_{\theta \in \mathcal{G}} L_S(\theta).} \tag{4}$$

This gives every category equal weight. The $S = 5$ and $S = 50$ experiments use the *same* candidate grid $\mathcal{G}$ and the same held-out folds for overlapping cases. The 5-image selections were frozen before the 50-image selections. The fixed tie tolerance was 10^{-10} in absolute loss, with grid order breaking ties.

Candidate grids and solver details

- **PnP-ADMM**: $\rho \in \{0.1, 0.25, 0.5, 1, 2\}$ and denoiser $\sigma \in \{0.03, 0.05, 0.075\}$, giving **15 candidates**. The solver permits 20–100 iterations, absolute residual tolerance 10^{-4}, relative residual tolerance 10^{-3}, and requires five consecutive passes to stop.
- **GS-PnP**: $\lambda \in \{0.01, 0.03, 0.1\}$, $\sigma \in \{0.03, 0.05, 0.075\}$, and independent initial step size $\tau_0 \in \{10, 100/3\}$, giving **18 candidates**. It allows 100 accepted iterations, 20 backtracking attempts, $\gamma = 0.1$, $\eta = 0.9$ and stopping tolerance 10^{-6}. The reconstruction output is the *paper-final* z image, not the internal x iterate [2].

The fidelity is $\frac{1}{2}\|A_\Theta x - y_\Theta\|_2^2$, with no additional division by the fit-set size. The operator audit confirmed the full and split Hadamard projections agree. The GS-PnP implementation initially showed GPU-dependent backtracking differences; after deterministic CUDA settings were enabled, the new adapter matched the frozen earlier GS solver exactly in the equivalence test.

Limits of the measurement-only criterion

The simulated PCM noise can be correlated across acquisition times. If the fit and held-out noise are correlated, the held-out prediction error is not necessarily an unbiased estimate of clean measurement prediction risk. Also, if $h \in \ker(A)$, then $A(x + h) = Ax$: unobserved fine detail cannot be assessed from acquired measurements alone. Finally, an SSDU fit fold contains fewer rows than the final 20% acquisition. The experiment keeps the fidelity convention fixed and transfers the selected parameter values directly to the full acquisition. These limitations must be considered when later comparing image metrics.

3 Step-by-step plan

Step	Action	Status
0	Verify the archived measurements; freeze five-image subset, 50-image set, candidate grid and 200 measurement-fold files.	Completed
1	Check normalisation, fit/holdout disjointness, numerical consistency and deterministic solver adapters.	Completed
2	Score all 15 ADMM and 18 GS-PnP candidates on five images, using four folds each.	Completed
3	Independently freeze PnP-ADMM-tuned-5 and GS-PnP-tuned-5, including all candidate scores.	Completed
4	Evaluate the identical candidate grids using all 50 validation images; reuse saved five-image folds.	Completed
5	Verify all 50-image candidates and freeze the two tuned-50 configurations.	Completed
6	Compare two tuned-5 SSDU methods and seven existing baselines on the fixed 450-image test acquisitions.	Not yet run
7	Produce combined PSNR, SSIM, HaarPSI, MSE, timing and paired-comparison figures.	Not yet run
8	Update this document with image quality results, failure analysis and reproducibility notes.	Future update

4 Experiment running

Data and reproducibility

The validation input is the **frozen acquisition-only archive** from the earlier oracle WLS experiment. It contains exactly **50 validation acquisitions and no test acquisitions**: 10 from each of CIGS, Organic, Perovskite, Reference and Silicone, under 20% sampling with NPL-default noise. The pilot image IDs are CIGS/0733, Organic/0008, Perovskite/0283, Reference/0768 and Silicone/0589 (validation permutation positions 901, 903, 905, 907 and 909). The source archive manifest begins with SHA256 `b30ff71a305bead9`.

Tuning ran on **tuni**. PnP-ADMM used the RTX 4070 and GS-PnP used the RTX 4090. Deterministic cuDNN and cuBLAS settings were enabled for numerical repeatability. The first five cases per grid were reused in full50 instead of recomputed. Across the 33 parameter configurations and 50 images with four folds each, the planned count was **6,600 unique fold reconstructions**. Each fold reconstruction can take many ADMM or GS iterations, and GS backtracking can require additional network evaluations.

The search was saved incrementally to JSON. Each candidate has per-image, per-fold losses, stopping diagnostics, aggregate category means and checksums. An unattended supervisor resumed the remaining full50 candidates after the initial single-candidate run, checked for incomplete candidates and failures, then generated frozen winner records. It finished at **13:42:43 UK time on 8 November 2026**. The numerical code revision recorded at run start was `c8842f5c42e8`; the grids, original acquisition manifest and code source files have hashes stored in the experiment archive.

Audit results

The fit and held-out projections matched their corresponding rows of the full operator in the numerical audit. Adjoint relative errors were 0 for the fit operator and about 9.4×10^{-8} for the held-out operator. Under deterministic settings, the GS adapter matched the previously frozen GS reconstruction exactly (maximum absolute difference 0). The ADMM adapter completed its iteration-bound smoke test. All **33 candidates** were completed in both the five-image and 50-image searches, with **no recorded candidate failures**. No ground-truth image or test-set acquisition was read for candidate selection.

5 Results and deliverables

Candidate ranking

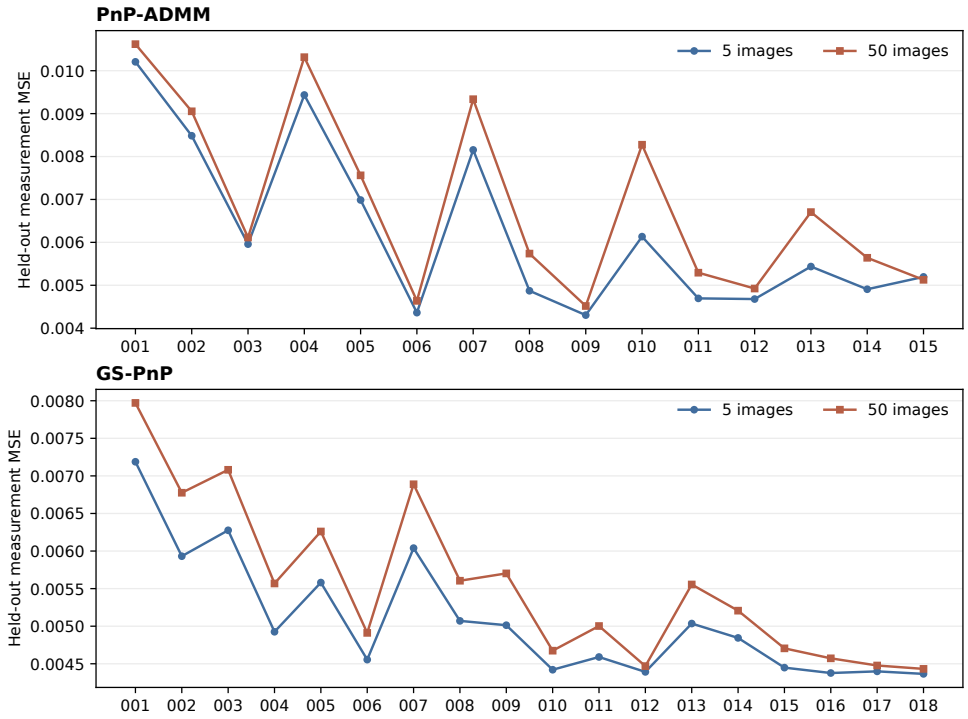


Figure 1: SSDU measurement-holdout losses over all candidate configurations. The two series use five and 50 validation images, respectively. Lower loss indicates better held-out measurement prediction, not necessarily higher PSNR.

Algorithm	Images	Three best candidates by SSDU loss
ADMM	5	`admm_009` (0.00430443), `admm_006` (0.00436202), `admm_012` (0.00467883)
ADMM	50	`admm_009` (0.00451722), `admm_006` (0.00463695), `admm_012` (0.00492435)
GS-PnP	5	`gs_018` (0.00436543), `gs_016` (0.00437685), `gs_012` (0.00439187)
GS-PnP	50	`gs_018` (0.00443193), `gs_012` (0.00446876), `gs_017` (0.00447703)

Table 2: The primary, ground-truth-free ranking uses equal-category means of four-fold held-out measurement MSE.

Per-category measurement error

	ADMM winner		GS-PnP winner	
Category	5 images	50 images	5 images	50 images
CIGS	0.003846	0.003747	0.003929	0.003773
Organic	0.004274	0.004375	0.004273	0.004319
Perovskite	0.004348	0.004085	0.004697	0.004135
Reference	0.004097	0.004871	0.004068	0.004770
Silicone	0.004957	0.005508	0.004860	0.005163

Table 3: Category-mean held-out measurement MSE for the selected configurations. The five-image stage has one image per category; the 50-image stage has ten.

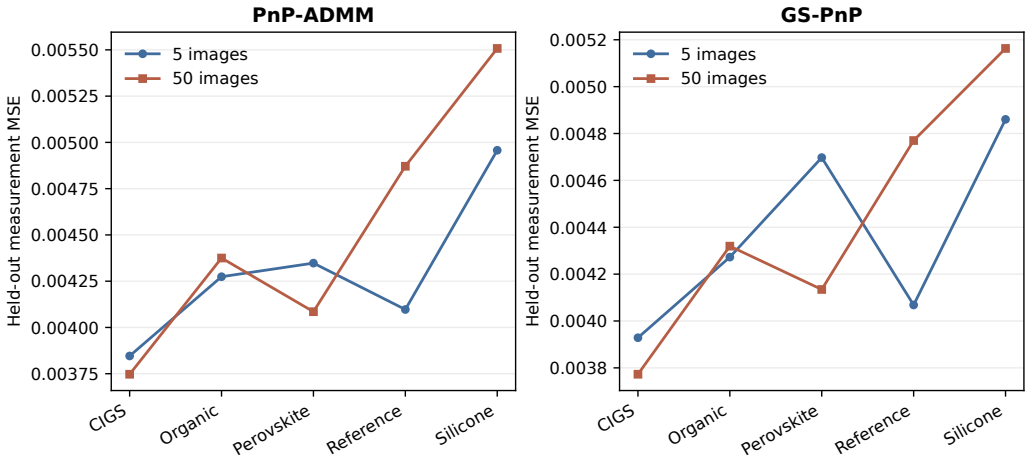


Figure 2: Variation of held-out prediction error across the five device categories, for the frozen winner at each validation-set size.

Deliverable files on tuni

The editable report `SSDU_PnP_Tuning_Report.tex`, the compiled `SSDU_PnP_Tuning_Report.pdf` and the two vector figures are kept together under the SSDU experiment's `report/` directory. The report can be edited and recompiled there after the 450-image evaluation has been incorporated.

The numerical evidence remains in the same experiment tree: `prepared/` contains acquisition and fold hashes, the grid and numerical audits; `pilot5/` and `full50/` contain all candidate scores and frozen parameter selections; `full50/runner/` contains execution logs. The Git branch `ssdu-pnp-tuning-20261108` stores the measurement-only scoring and reconstruction code. The intended 11-method test comparison is recorded in `comparison_methods.json`. No source images or archived acquisitions were modified.

Main 450-image comparison: tuned-5 only

The 5-image and 50-image SSDU selections chose the *same* parameter configuration for each PnP algorithm. Their 450-image test metrics are therefore identical. To avoid duplicated results in the primary comparison, we retain only *PnP-ADMM-tuned-5* and *GS-PnP-tuned-5* alongside the seven original baselines. The separately frozen tuned-50 selections and the previous full 11-label comparison are preserved in the version-2 archive.

All methods use the exact same fixed 20% NPL-noisy test acquisition per image, with all 450 measurement and mask SHA256 hashes checked. The supervised validation-tuned Step-6 GS-PnP oracle uses its *paper-final* z output. No test-based parameter selection was performed.

Method	PSNR (dB)	SSIM	HaarPSI	MSE
Zero-filled	24.97 ± 1.97	0.4823 ± 0.0527	0.5513 ± 0.0520	3.526e-03 ± 1.623e-03
SPGL1-BP	25.42 ± 1.53	0.4353 ± 0.0354	0.6452 ± 0.0772	3.061e-03 ± 1.184e-03
SPGL1-BPDN	29.37 ± 3.41	0.7161 ± 0.1036	0.6581 ± 0.0994	1.552e-03 ± 1.185e-03
PnP-ADMM oracle	35.75 ± 3.43	0.8970 ± 0.0770	0.8814 ± 0.0619	3.536e-04 ± 2.433e-04
GS-PnP oracle	35.11 ± 3.64	0.8709 ± 0.0871	0.8628 ± 0.0735	4.184e-04 ± 2.866e-04
TV	30.03 ± 1.94	0.7212 ± 0.0553	0.7844 ± 0.0504	1.103e-03 ± 5.472e-04
UTV-C60	32.30 ± 2.93	0.8450 ± 0.0882	0.7699 ± 0.1101	7.279e-04 ± 4.532e-04
PnP-ADMM tuned-5	35.75 ± 3.43	0.8970 ± 0.0770	0.8814 ± 0.0619	3.536e-04 ± 2.433e-04
GS-PnP tuned-5	32.59 ± 2.70	0.7570 ± 0.0967	0.8208 ± 0.0775	6.628e-04 ± 3.910e-04

Table 4: 450-image mean ± sample standard deviation on raw unclipped reconstructions, for nine retained method labels.

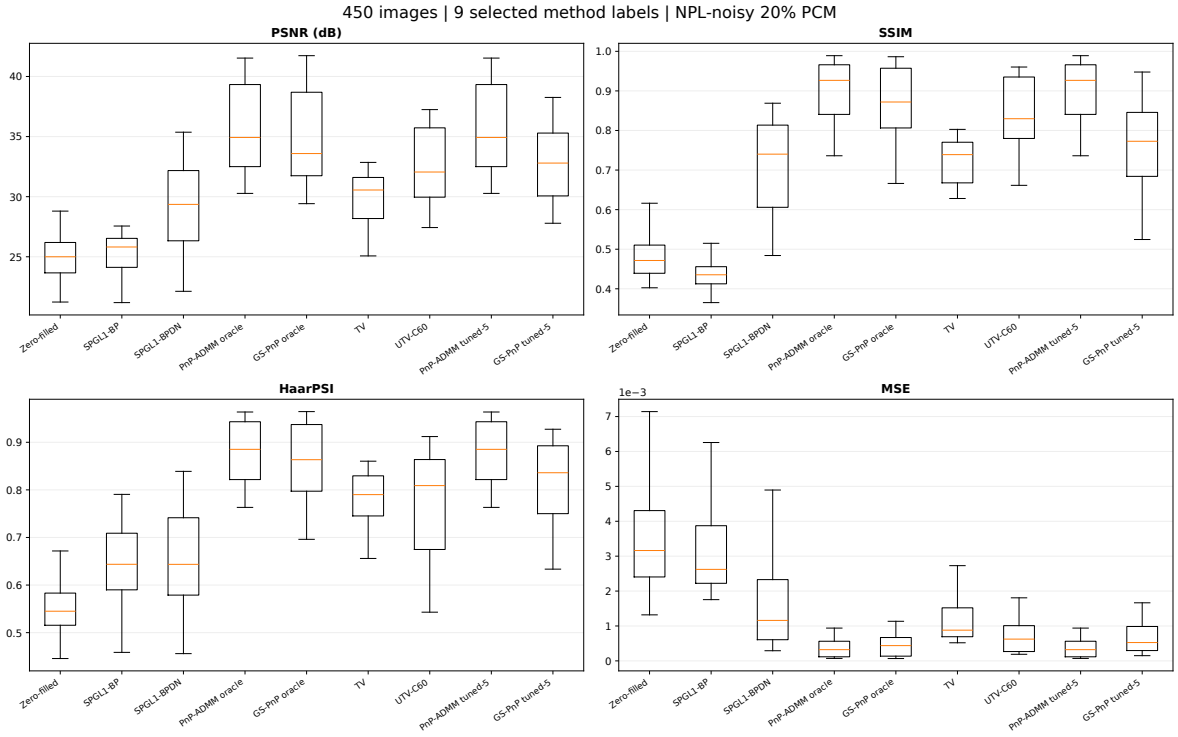


Figure 3: Box plots of the paired 450-image PSNR, SSIM, HaarPSI and MSE results for the seven baselines and two tuned-5 methods.

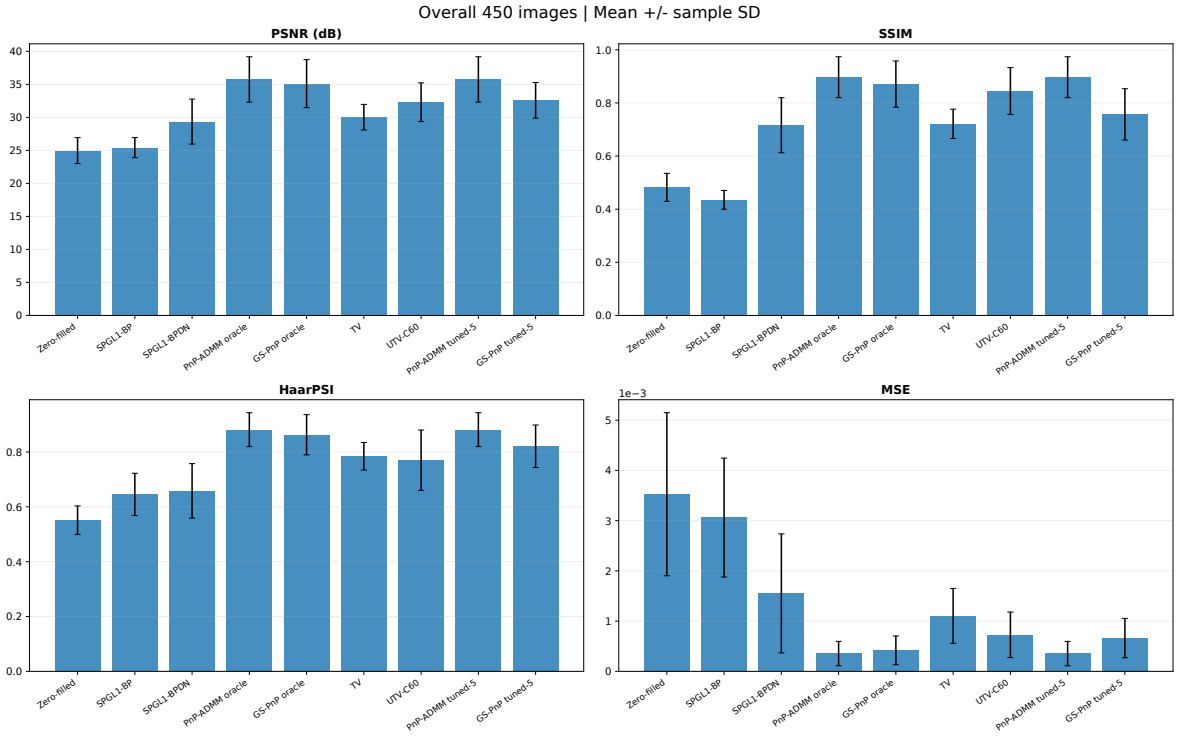


Figure 4: Overall 450-image mean and sample standard deviation for the nine retained method labels.

What does agreement with the oracle mean?

PnP-ADMM: SSDU-tuned-5 chooses exactly the same parameters ($\rho = 0.5, \sigma = 0.075$) and stopping rule as the earlier supervised validation-tuned ADMM benchmark. Therefore both methods produce the same 450-image scores: mean PSNR 35.754 dB, SSIM 0.8970 and HaarPSI 0.8814. This shows that SSDU reaches *that*

oracle-selected benchmark. It does *not* prove that these hyperparameters are globally optimal over untested configurations.

GS-PnP: SSDU-tuned-5 gives mean PSNR 32.590 dB, compared with 35.110 dB for the frozen supervised-validation Step-6 GS-PnP oracle. The paired PSNR difference is -2.519 dB, with sample SD 1.890 dB. Thus the current measurement-only SSDU criterion did not recover the same image-space quality for GS-PnP. This is an observed mismatch, not proof of a specific cause.

Device-category scores

Category	ZF	SPGL1	BPDN	ADMM oracle	GS oracle	TV	C60	ADMM-5	GS-5
CIGS	25.71	26.49	31.10	39.30	37.89	31.16	33.04	39.30	35.31
Organic	25.49	26.37	32.05	38.90	38.94	31.50	35.72	38.90	34.96
Perovskite	27.06	26.43	31.60	35.48	35.35	31.33	32.61	35.48	31.70
Reference	24.25	24.39	26.72	32.14	31.72	28.30	29.63	32.14	29.73
Silicone	22.34	23.44	25.38	32.95	31.65	27.87	30.49	32.95	31.25

Table 5: Per-category mean PSNR (dB) over 90 test images per category. The complete 450-image row-level metrics are retained separately as CSV.

A separate five-page PDF provides the per-category means and standard deviations for all nine method labels. It is saved with the experiment's comparison figures.

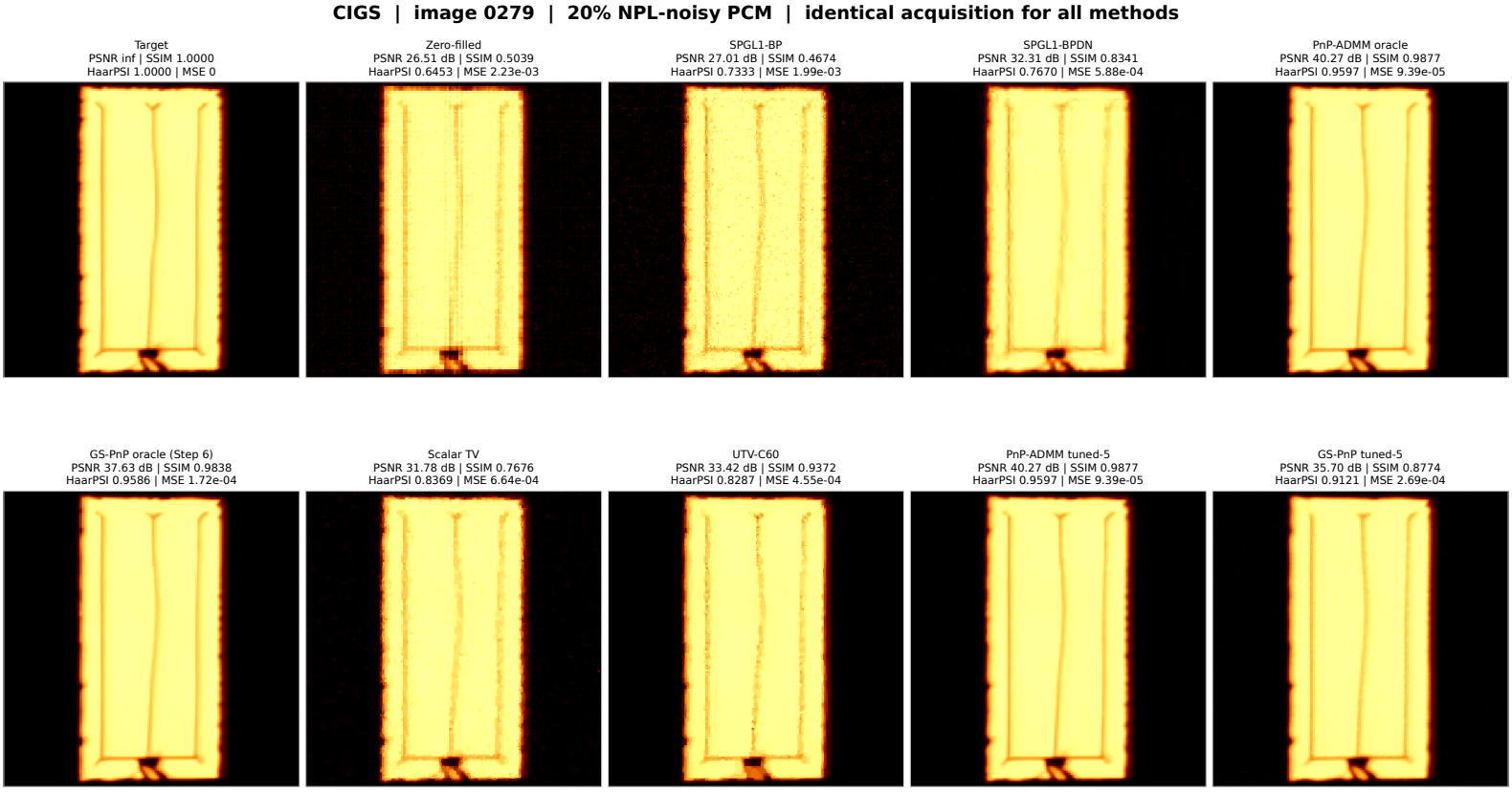
Five visual comparisons with per-panel metrics

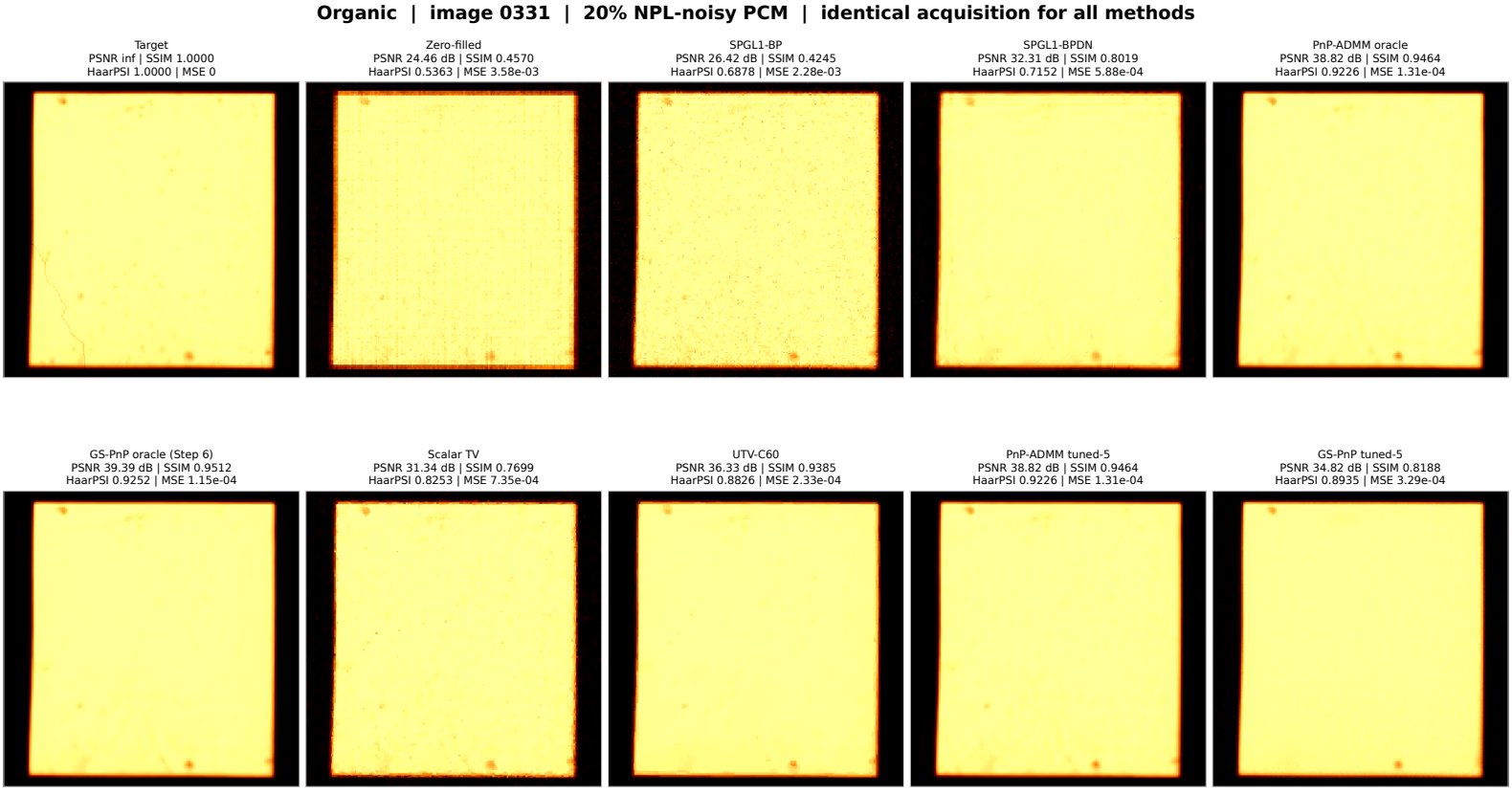
We fixed one case per device category before viewing reconstruction scores. CIGS image 0279 was requested explicitly; the other four were drawn with a recorded fixed random seed from the ten already preselected 450-test visualisation cases per category:

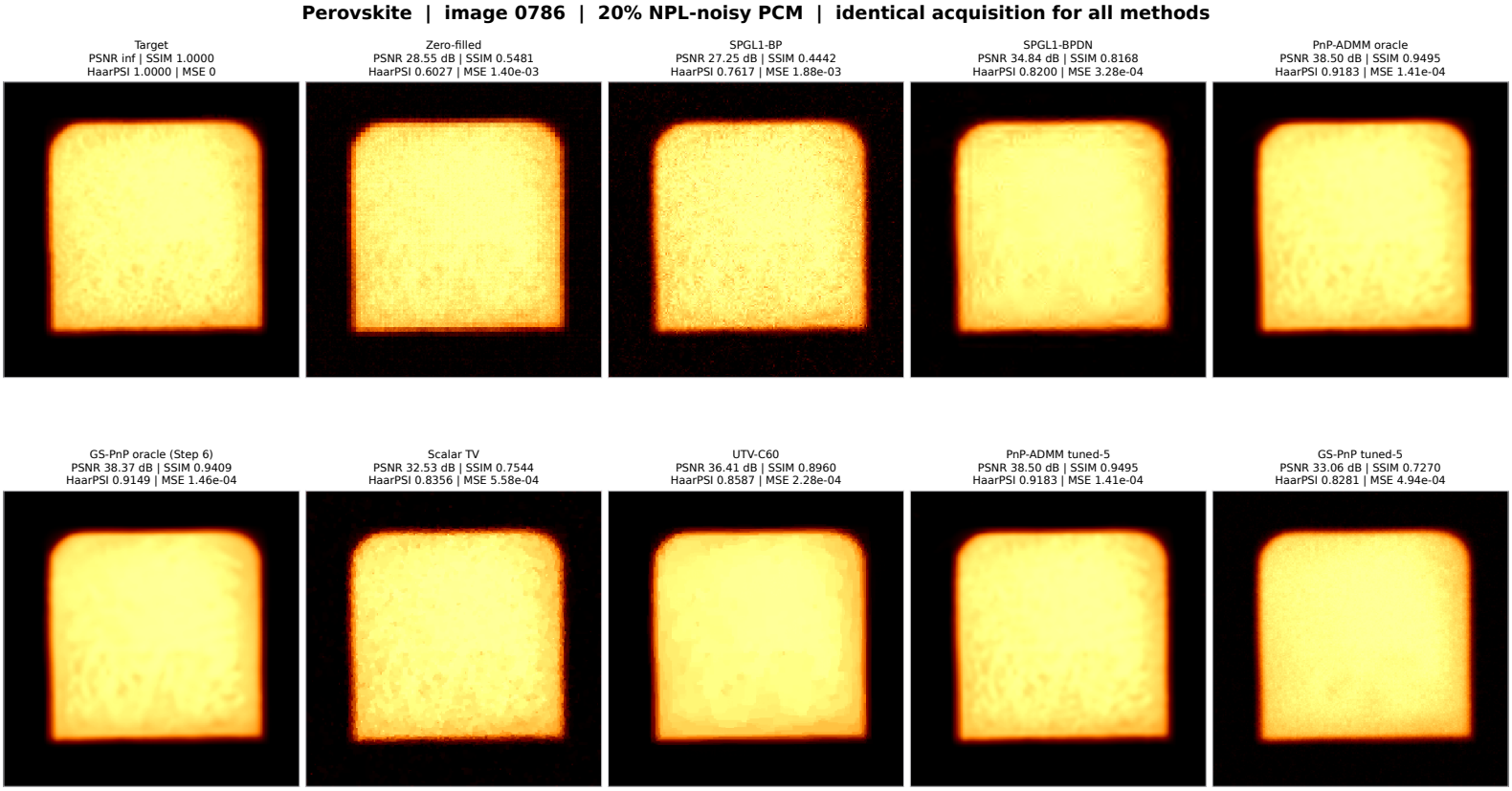
Category	Image ID
CIGS	0279 (specified)
Organic	0331
Perovskite	0786
Reference	0279
Silicone	0370

Each page shows ground truth and all nine retained methods, in a two-by-five panel layout. Every reconstruction uses the same stored 20% noisy PCM acquisition. The display uses AFM-hot capped at 0.8, with fixed image intensity $[0, 1]$. Below each method label we give PSNR, SSIM, HaarPSI and MSE, computed on *raw* image values. The historical BP, BPDN, TV, C60 and ADMM examples were replayed against their stored metrics; all numerical checks passed.

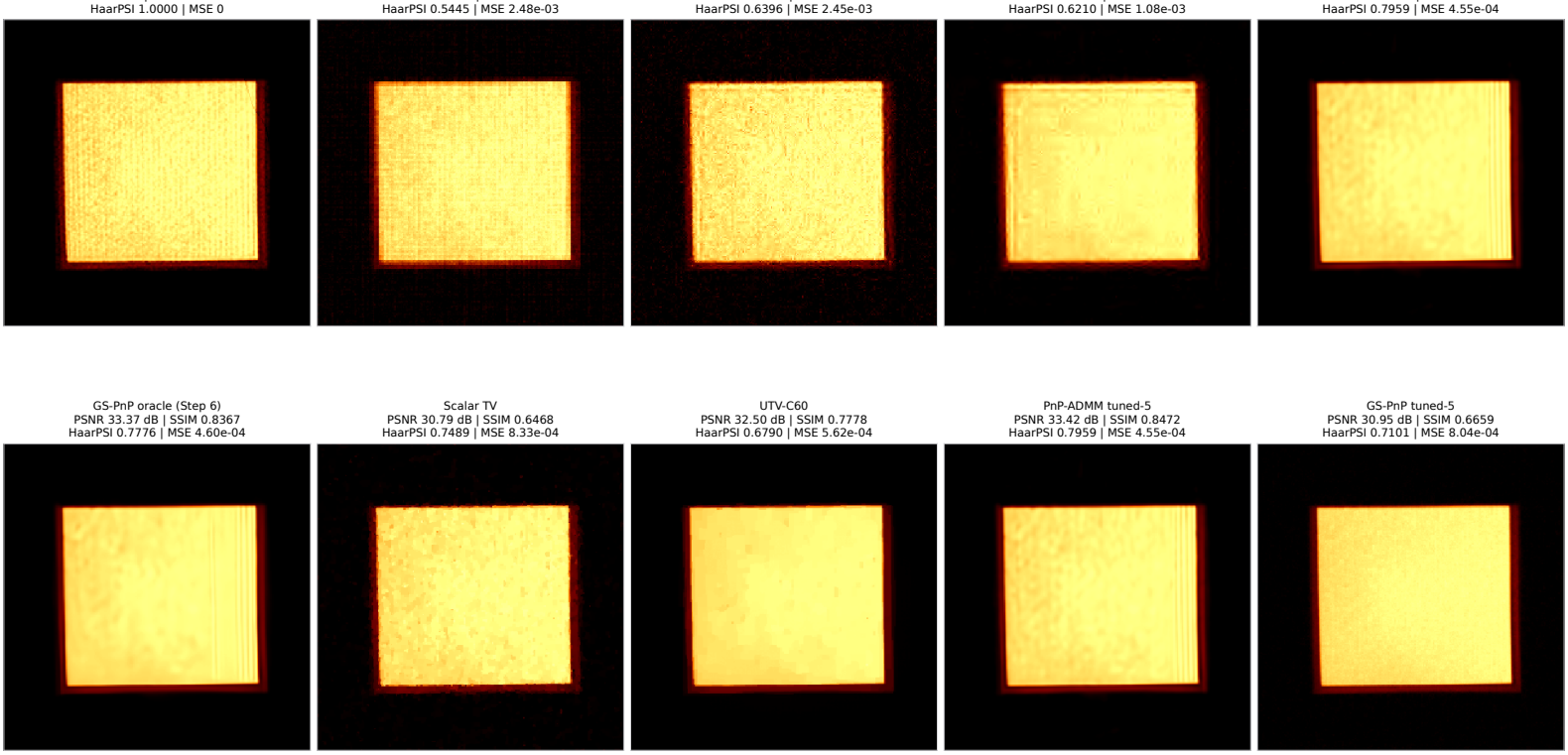
The five landscape comparison pages are embedded immediately below. They also remain in a standalone PDF for detailed inspection.


CIGS | image 0279 | 20% NPL-noisy PCM | identical acquisition for all methods
Target
PSNR inf | SSIM 1.0000
HaarPSI 1.0000 | MSE 0
Zero-filled
PSNR 26.51 dB | SSIM 0.5039
HaarPSI 0.6453 | MSE 2.23e-03
SPGL1-BP
PSNR 27.01 dB | SSIM 0.4674
HaarPSI 0.7333 | MSE 1.99e-03
SPGL1-BPDN
PSNR 32.31 dB | SSIM 0.8341
HaarPSI 0.7670 | MSE 5.88e-04
PnP-ADMM oracle
PSNR 40.27 dB | SSIM 0.9877
HaarPSI 0.9597 | MSE 9.39e-05
GS-PnP oracle (Step 6)
PSNR 37.63 dB | SSIM 0.9838
HaarPSI 0.9586 | MSE 1.72e-04
Scalar TV
PSNR 31.78 dB | SSIM 0.7676
HaarPSI 0.8369 | MSE 6.64e-04
UTV-C60
PSNR 33.42 dB | SSIM 0.9372
HaarPSI 0.8287 | MSE 4.55e-04
PnP-ADMM tuned-5
PSNR 40.27 dB | SSIM 0.9877
HaarPSI 0.9597 | MSE 9.39e-05
GS-PnP tuned-5
PSNR 35.70 dB | SSIM 0.8774
HaarPSI 0.9121 | MSE 2.69e-04
1/5 | AFM-hot capped at 0.8; fixed intensity [0,1]. The ADMM oracle and tuned-5 panels are identical by configuration.

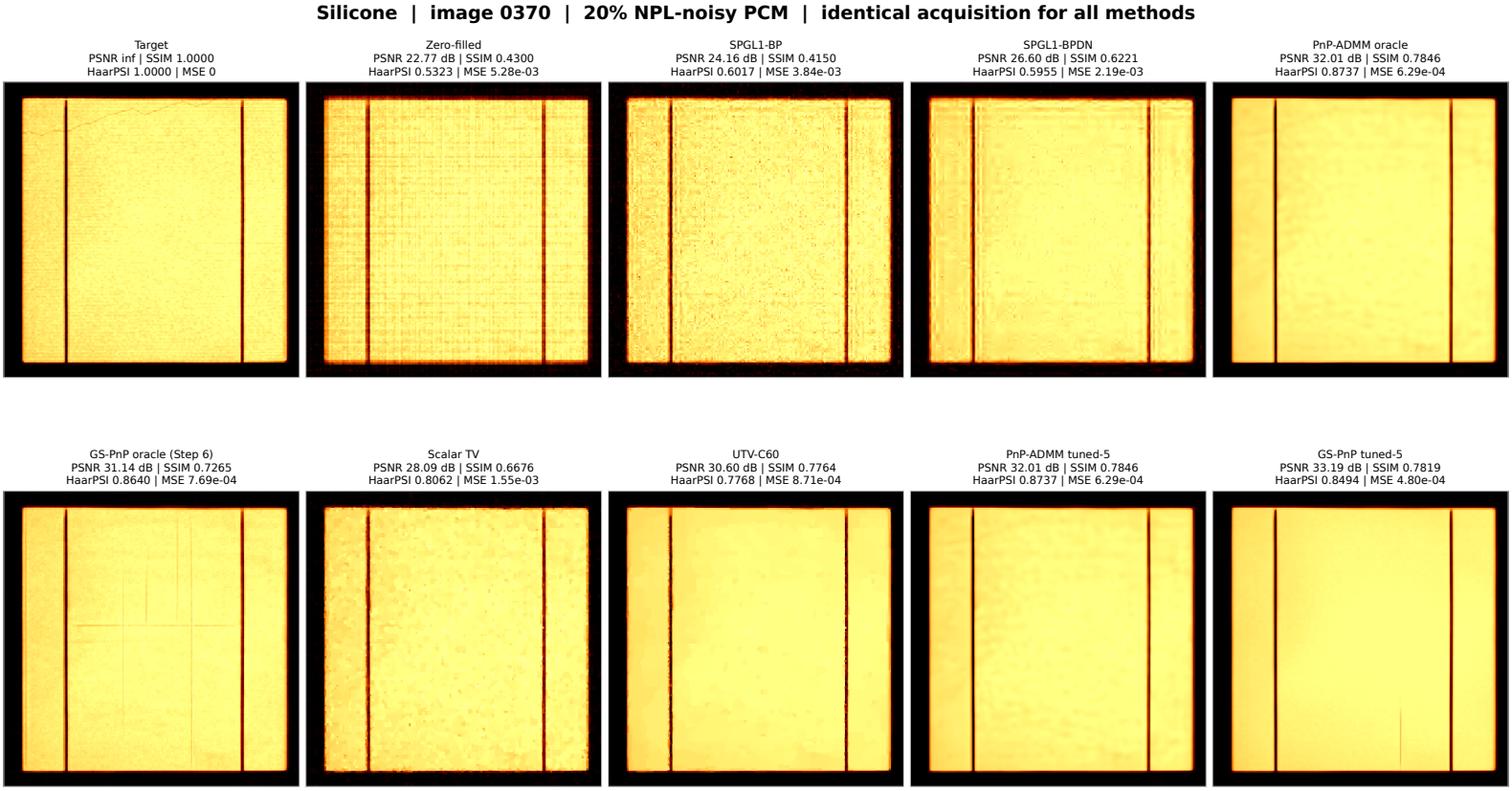

Organic | image 0331 | 20% NPL-noisy PCM | identical acquisition for all methods
Target
PSNR inf | SSIM 1.0000
HaarPSI 1.0000 | MSE 0
Zero-filled
PSNR 24.46 dB | SSIM 0.4570
HaarPSI 0.5363 | MSE 3.58e-03
SPGL1-BP
PSNR 26.42 dB | SSIM 0.4245
HaarPSI 0.6878 | MSE 2.28e-03
SPGL1-BPDN
PSNR 32.31 dB | SSIM 0.8019
HaarPSI 0.7152 | MSE 5.88e-04
PnP-ADMM oracle
PSNR 38.82 dB | SSIM 0.9464
HaarPSI 0.9226 | MSE 1.31e-04
GS-PnP oracle (Step 6)
PSNR 39.39 dB | SSIM 0.9512
HaarPSI 0.9252 | MSE 1.15e-04
Scalar TV
PSNR 31.34 dB | SSIM 0.7699
HaarPSI 0.8253 | MSE 7.35e-04
UTV-C60
PSNR 36.33 dB | SSIM 0.9385
HaarPSI 0.8826 | MSE 2.33e-04
PnP-ADMM tuned-5
PSNR 38.82 dB | SSIM 0.9464
HaarPSI 0.9226 | MSE 1.31e-04
GS-PnP tuned-5
PSNR 34.82 dB | SSIM 0.8188
HaarPSI 0.8935 | MSE 3.29e-04
2/5 | AFM-hot capped at 0.8; fixed intensity [0,1]. The ADMM oracle and tuned-5 panels are identical by configuration.


Perovskite | image 0786 | 20% NPL-noisy PCM | identical acquisition for all methods
Target
PSNR inf | SSIM 1.0000
HaarPSI 1.0000 | MSE 0
Zero-filled
PSNR 28.55 dB | SSIM 0.5481
HaarPSI 0.6027 | MSE 1.40e-03
SPGL1-BP
PSNR 27.25 dB | SSIM 0.4442
HaarPSI 0.7617 | MSE 1.88e-03
SPGL1-BPDN
PSNR 34.84 dB | SSIM 0.8168
HaarPSI 0.8200 | MSE 3.28e-04
PnP-ADMM oracle
PSNR 38.50 dB | SSIM 0.9495
HaarPSI 0.9183 | MSE 1.41e-04
GS-PnP oracle (Step 6)
PSNR 38.37 dB | SSIM 0.9409
HaarPSI 0.9149 | MSE 1.46e-04
Scalar TV
PSNR 32.53 dB | SSIM 0.7544
HaarPSI 0.8356 | MSE 5.58e-04
UTV-C60
PSNR 36.41 dB | SSIM 0.8960
HaarPSI 0.8587 | MSE 2.28e-04
PnP-ADMM tuned-5
PSNR 38.50 dB | SSIM 0.9495
HaarPSI 0.9183 | MSE 1.41e-04
GS-PnP tuned-5
PSNR 33.06 dB | SSIM 0.7270
HaarPSI 0.8281 | MSE 4.94e-04
3/5 | AFM-hot capped at 0.8; fixed intensity [0,1]. The ADMM oracle and tuned-5 panels are identical by configuration.

Reference | image 0279 | 20% NPL-noisy PCM | identical acquisition for all methods



4/5 | AFM-hot capped at 0.8; fixed intensity [0,1]. The ADMM oracle and tuned-5 panels are identical by configuration.



5/5 | AFM-hot capped at 0.8; fixed intensity [0,1]. The ADMM oracle and tuned-5 panels are identical by configuration.

Next GS-PnP study: expanded blind SSDU search (planned)

The original SSDU GS-PnP winner was at the *maximum* of the candidate grid along *all three* parameter dimensions: $\lambda = 0.1$, denoiser $\sigma = 0.075$ and $\tau_0 = 100/3$. Therefore the original parameter search did not demonstrate an interior minimum. We have predeclared a wider Cartesian grid:

Parameter	Candidate values
λ	0.10, 0.20, 0.40
Denoiser σ	0.075, 0.10, 0.15
Initial τ_0	33.3333, 50, 75

The $3 \times 3 \times 3 = 27$ triples include the old winner and 26 previously untested combinations. Apply the exact original five validation images and four held-out folds each: 540 GS-PnP reconstruction calls. Keep the DRUNet model, loss function, operator normalisation and GS stopping/backtracking rules unchanged. Independently vary λ, σ and τ_0.

Before running, record all candidates and numerical instability policies. Inspect accepted step sizes, $\lambda\tau$, line-search rejections, stopping reason and held-out MSE. Select the winner from *validation measurement loss only*, freeze it, and report whether the new minimum again lies on an upper boundary.

The extended sweep is a **future experiment and has NOT been run yet**. Increasing parameter magnitudes does not necessarily improve PSNR or preserve details, particularly when PCM measurement noise is correlated. Further evaluation on the previously inspected 450 test cases must be labelled retrospective and cannot be used to repeatedly tune parameters.

6 Conclusion

The SSDU-style measurement-only tuning study is complete. Both 5-image and 50-image validation sets selected the same parameter configuration within each PnP algorithm's *tested grid*. Thus five images were sufficient for selection stability in this particular protocol. We retain the tuned-5 configurations for subsequent comparisons, while preserving all tuned-50 results and selections in the original study archive.

For PnP-ADMM, ground-truth-free tuned-5 matched the old supervised-validation-tuned benchmark *exactly*: both use $\rho = 0.5$, $\sigma = 0.075$ and the same stopping criterion, yielding mean PSNR 35.754 dB over 450 matched test acquisitions. This is agreement with the previous benchmark, not proof of globally optimal reconstruction parameters.

For GS-PnP, tuned-5 gives mean PSNR 32.590 dB, compared with 35.110 dB for the Step-6 supervised-validation oracle (paired difference -2.519 dB). The selected λ, denoiser σ and initial τ_0 are all at the top of the original candidate grid, so the grid should be expanded using *validation-only* SSDU scoring. The proposed 27-configuration higher-value sweep is written and frozen but has *not* been run.

The five case comparisons with raw-image metrics give qualitative context; they cannot establish the preservation of thin lines without further targeted evaluation. Temporally correlated noise and the undersampled Hadamard nullspace may limit how well SSDU prediction loss tracks image quality. Since these 450 test images have already been inspected in related research, the evaluation is retrospective.

Primary references

[1] B. Yaman et al., "Self-Supervised Learning of Physics-Guided Reconstruction Neural Networks without Fully-Sampled Reference Data", *Magnetic Resonance in Medicine*, 84:3172–3191, 2020. See the method for partitioning acquired measurements into data-consistency and loss sets. https://doi.org/10.1002/mrm.28378.

[2] S. Hurault, A. Leclaire and N. Papadakis, "Gradient Step Denoiser for Convergent Plug-and-Play", *ICLR*, 2022. See Algorithm 1 and the backtracking procedure. https://openreview.net/pdf?id=fPhKeld3Okz.